

National University of Computer and Emerging Sciences, Lahore Campus



Course: Analog and Digital Communications
 Program: BS(Electrical Engineering)
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 Roll No: 15-4522

Instruction/Notes: Attempt all parts of a question together.

Question # 1 [CLO 5]

[15]

- State the sampling theorem.
- A signal has a bandwidth of 5.5MHz. Determine the sampling rate when sampling is being done at double the Nyquist rate.
- State the practical difficulties in signal reconstruction having gone through practical sampling.

Question # 2 [CLO 4]

[15]

The scheme shown in Figure 2(a) is used for angle modulation, where voltage controlled oscillator (VCO) frequency $f_0 = 100$ MHz, VCO amplitude = 1V peak, $k_f = 2000\pi$ and $k_p = \frac{\pi}{2}$. Modulating signal $m(t)$ is given in Figure 2(b). Consider the essential bandwidth of the signal $m(t)$ to be the frequency corresponding to the fifth harmonic.

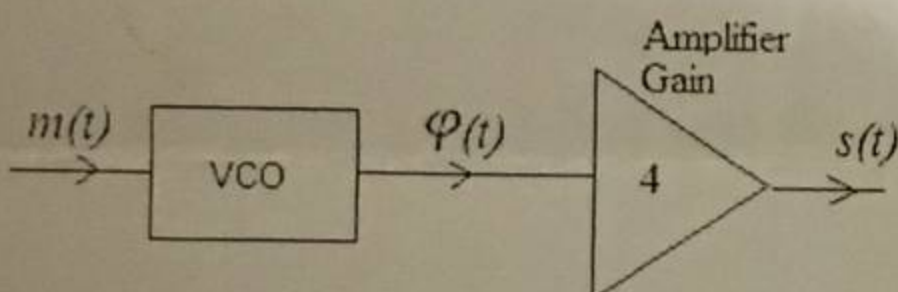


Figure 2(a)

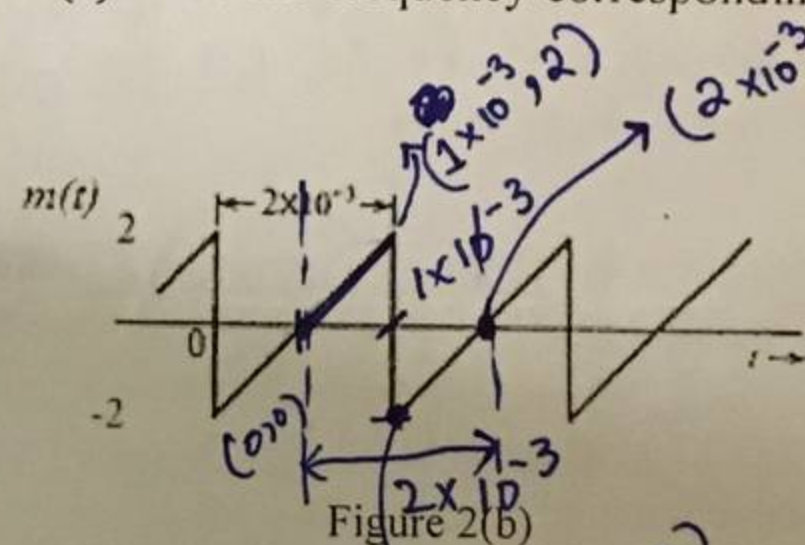


Figure 2(b)

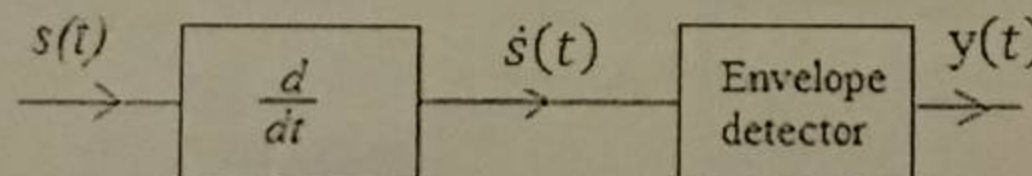


Figure 2(c)

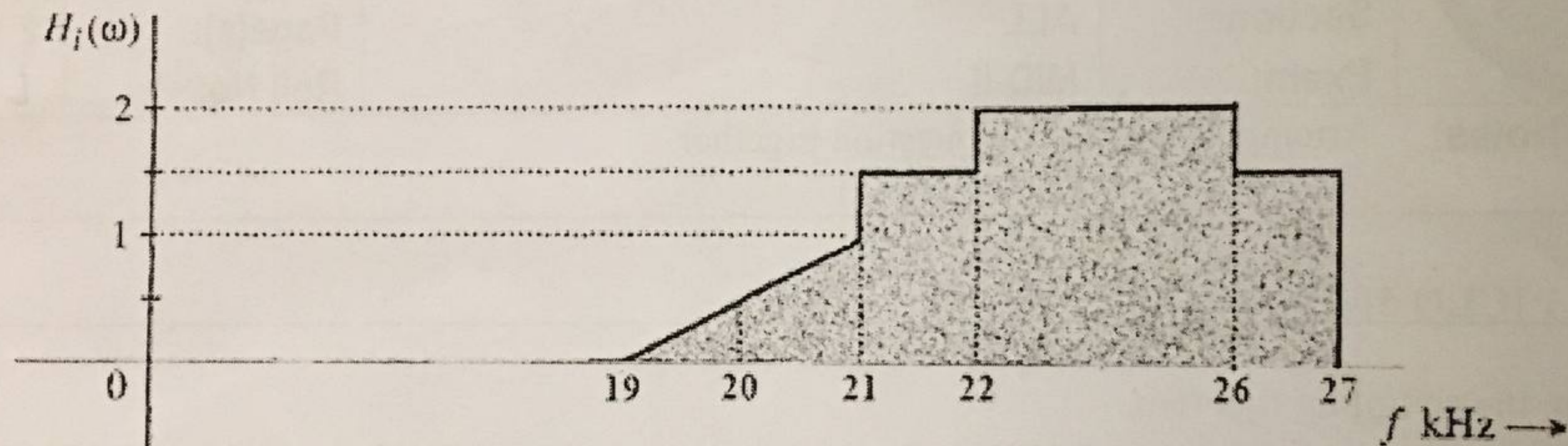
- Write equations for signals $\phi(t)$ and $s(t)$.
- Determine the frequency deviation Δf .
- Determine the power and bandwidth of angle modulated signal $s(t)$.
- If $s(t)$ is demodulated by a differentiator followed by an envelope detector as shown in Figure 2(c), write equations for $\dot{s}(t)$ and $y(t)$.

Hint: Use the indefinite integral of $m(t)$; that is, take the value of the integral at $t = -\infty$ to be 0.

Question # 3 [CLO 3]

[15]

Consider a VSB amplitude modulation system. The baseband signal bandwidth is 6KHz. The carrier frequency is $f_c = 20\text{KHz}$. Suppose that the vestigial shaping filter $H_i(\omega)$ at transmitter end has a frequency response as shown in Figure below



- Find the bandwidth of this transmission.
- Sketch the necessary equalizer filter response $H_o(\omega)$ for distortion-less reception.
- Design and illustrate a receiver system block diagram for VSB.